ENGLISH WORKSHEET / BLANK RESPONSE COPY

Strawberry DNA extraction

Student worksheet · Teacher-supervised practical · Record your own evidence.

NAME	CLASS	DATE
_____	_____	_____

Learning objectives

By the end of the practical, I can…

01	Explain how membrane disruption and precipitation help make DNA-rich fibres visible.
02	Describe what mashing, detergent, salt, filtration and cold ethanol do.
03	Follow the method safely and record observations without claiming the crude extract is pure DNA.

Risk assessments & controls — read before the method and observation record

Check the control measures with your teacher before handling any material.

MATERIAL / HAZARD	RISK	CONTROL MEASURE
Cold ethanol — flammable liquid / vapour	Vapour may irritate; flames can ignite vapour.	Teacher-directed use only. Keep capped, cold and away from flames or other ignition sources.
Detergent — eye / skin irritant	A splash can irritate eyes or skin.	Wear goggles. Avoid contact; rinse splashes with plenty of water and tell the teacher.
Glass and spills	Breakage can cut; liquid can make floors slippery.	Check glassware. Report breakage and clean spills promptly using the class procedure.

[ ] Goggles on	[ ] Bench clear	[ ] Teacher's instructions checked

Before you begin: Which control measure will you check first, and why?

01 / BEFORE THE PRACTICAL

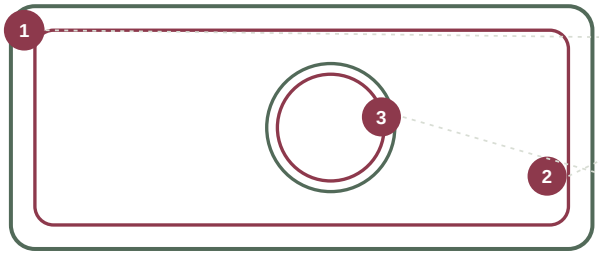
Before the practical

Think first; return to your predictions after the experiment.

1.1 Cultivated garden strawberries are octoploid (8n). In your own words, what does 8n tell you about chromosome sets? Does ploidy alone tell you the total DNA mass in a cell? Explain.

1.2 Use the word bank to label the numbered barriers in the simplified cell sketch.

WORD BANK: cell wall / cell membrane / nuclear envelope



1. _____
2. _____
3. _____

SIMPLIFIED CELL SKETCH - NOT TO SCALE

1.3 Before adding the cold liquid, predict where the pale fibres might appear and describe what you expect to see.

02 / METHOD IN PICTURES

Method in pictures

Follow the steps in order. Keep the filtrate and ethanol layers separate.

PREPARE THE BUFFER	GET THE EQUIPMENT READY
Mix 90 cm³ water + 10 cm³ detergent + 3 g salt. Stir until the salt dissolves. Add 10 cm³ of prepared buffer to the mashed fruit.	One hulled strawberry; sealable bag; gauze-lined funnel; clean tube; clean glass rod; goggles; ice-cold ethanol.



01-02 / MASH FRUIT + BUFFER

Put one hulled strawberry in a sealable bag and mash gently for 2 minutes. Stir the buffer until the salt dissolves, then add 10 cm³ to the fruit.

[ ] DONE



03 / FILTER THE MIXTURE

Pour slowly through gauze in a funnel into a clean tube. Collect the filtrate; leave most visible pulp on the gauze.

[ ] DONE



04 / LAYER COLD ETHANOL

Measure 5 cm³ filtrate and an equal 5 cm³ volume of ice-cold ethanol. Run ethanol slowly down the tube wall; do not shake or stir.

[ ] DONE



05 / SPOOL THE FIBRES

Touch a clean glass rod to the boundary between liquid layers. Turn slowly in one direction to collect any pale, stringy material.

[ ] DONE

WHAT TO LOOK FOR / Pale, stringy DNA-rich material may appear at the boundary after ice-cold ethanol meets the filtrate. The extract is crude, not pure DNA.

03 / OBSERVATION RECORD

Observation record

Describe what you see before explaining why it happened. Sketch or write; use the full space.

STAGE	WHAT I SEE / SKETCH OR DESCRIBE
After mashing and adding buffer	
After filtration: liquid and pulp retained on the gauze	
Where ethanol meets the filtrate	
After spooling: colour, texture and location	

3.1 Which observation is clearest or most useful as evidence? Describe it without explaining it yet.

3.2 What can the appearance of fibres show—and what cannot it prove about the crude extract?

04 / EXPLAIN AND EVALUATE

Explain and evaluate

Use evidence from your sample. A visible result is an observation, not proof of purity.

WHAT DOES EACH STEP OR MATERIAL DO? Write an explanation in your own words.

STEP / MATERIAL	MY EXPLANATION
Mashing	
Detergent	
Salt	
Filtration	
Ice-cold ethanol	
Spooling with a glass rod	

FILTRATION CHECK / Tick one destination for each item.

MATERIAL	RETAINED BY GAUZE	IN THE FILTRATE
Coarse pulp and large cell-wall fragments	[ ]	[ ]
Unmashed strawberry tissue	[ ]	[ ]
Soluble DNA	[ ]	[ ]
Water, salt ions and detergent molecules	[ ]	[ ]

4.1 Why is salt added to the buffer? Include DNA's charge in your explanation.

4.2 Why use ice-cold ethanol and add it gently without mixing the layers?